

Fig. 1: The mechanism of charge transfer and transport in a bulk-heterojunction organic PV structure

still being developed by technology incubators and start-up companies. Thus today photovoltaics market is dominated by silicon-photovoltaics.

The two metrics for evaluating and comparing different photovoltaic technologies, in conjunction with their power conversion efficiency, include:

1. Levelised cost of energy (LCOE) = Total life-cycle cost/energy produced

2. Energy payback time (EPBT) = Energy consumed over the life of the module/Energy generated over its lifetime

Fortunately, these two metrics predict bright future for the new emerging technologies.

Conventional silicon cells require ultra-high-purity silicon—of the order of 99,999 per cent pure. These cells are made using energy-intensive crystal growth and vapour deposition methods. One of the drawbacks of silicon technology that further increases its cost and limits applications is that silicon is indirect band semiconductor, so it doesn't absorb sunlight strongly. Therefore relatively thick layer of silicon is used in cells, which become brittle and hence have to be supported on a rigid, heavy piece of glass. A silicon solar cell requires 14 steps to manufacture, including preparations requiring the use of high-heat, expensive automation and clean rooms.

In comparison, the new emerging photovoltaic cells are made from low-cost materials, including organic poly-

mers, small molecules and various types of inorganic compounds, and can be fabricated on flexible supports via inexpensive solution-phase techniques common in plastics manufacturing, such as high-speed roll-to-roll printing. In addition, these PV cells use 1000 times less, light-absorbing material than silicon solar cells, which further reduces their cost.

### Third-generation solar cells

**Organic photovoltaics.** In organic photovoltaics, bulk heterojunction structure (a mixture of donor and acceptor materials in a bulk device) is most commonly used due to its high efficiency and compatibility with low-cost manufacturing techniques such as inkjet printing, spin coating and roller casting. These PVs are renowned as high-PCE polymer-fullerene solar cells.

When the light incident on an organic solar cell passes through the transparent anode to the donor, the photon is absorbed at the donor layer

(based on the generalisation that the organic donor is the light-absorbing material). The charge carriers are generated by photo-induced electron transfer. An electron is excited from the highest occupied molecular orbital (HOMO) to the lowest unoccupied molecular orbital (LUMO), creating a bound electron-hole pair known as an exciton. The exciton diffuses to the donor-acceptor interface. There, the excited electron is transferred to the LUMO of the acceptor, and the energy difference between the LUMOs of the donor and acceptor drives dissociation of the exciton into free charge carriers. The mechanism is depicted in Fig. 1.

For efficient solar cells, the created charge carriers need to be transported to corresponding electrodes within their respective lifetimes. The charge carriers are finally extracted through ITO-coated glass matching the HOMO level of conjugated polymer (hole contact) used on illumination side and metal electrode matching the LUMO of fullerene PCBM (electron contact).

Organic PV is an emerging solar technology with improving cell efficiency (currently 13.2 per cent), boosting initial lifetime (more than 5000 hours) and potential for roll-to-roll manufacturing processes. It is well suited for building-integrated PV market because of the availability of absorbers in various colours and the ability to make efficient transparent devices.

It is heartening to find that several printing and coating technologies—such as gravure print-

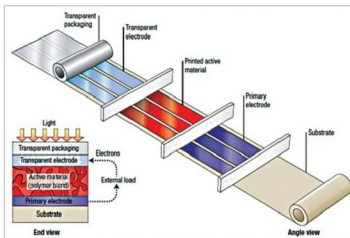


Fig. 2: Polymer-based photovoltaic cells can be manufactured using standard printing processes